DATASHEET

# AprilGrid Calibration Target, Aluminum Composite

AprilGrid uses AprilTag 36h11 markers arranged in a regular grid. It is the standard target for Kalibr camera–IMU calibration and is also supported by the OpenCV ArUco module (`cv2.aruco.DICT_APRILTAG_36h11`). Each tag carries a unique ID, so partially visible boards still yield usable detections — useful at field edges where distortion is greatest.

## 1 Ordering

| Part number | Size (W × H) | Tags (rows × cols) | Tag / gap | Markers |
|---|---|---|---|---|
| CT-APR-P-200X150-T36 | 200 × 150 mm | 4 × 6 | 25 / 7.5 mm | 24 |
| CT-APR-P-300X200-T36 | 300 × 200 mm | 5 × 7 | 30 / 9 mm | 35 |
| CT-APR-P-400X300-T36 | 400 × 300 mm | 6 × 8 | 35 / 10.5 mm | 48 |
| CT-APR-P-600X400-T36 | 600 × 400 mm | 7 × 10 | 45 / 13.5 mm | 70 |
| CT-APR-P-800X600-T36 | 800 × 600 mm | 9 × 12 | 50 / 15 mm | 108 |

## 2 Key specifications

ACCURACY

±0.3 mm + 0.2 mm/m (any tag center to nominal grid, at 20 °C)

SUBSTRATE

Aluminum / LDPE / aluminum composite, 6 mm

FLATNESS

≤ 0.5 mm per meter, board supported on three points

Unless stated otherwise, dimensions are nominal at 20 °C.

## 1 Specifications · geometry

| FEATURE | SPECIFICATION |
|---|---|
| Feature position tolerance | ±0.3 mm + 0.2 mm/m (any tag center to nominal grid, at 20 °C) |
| Tag pitch, mean | ±0.05 mm |
| Tag-to-gap ratio | 0.3 |
| Edge radius | R6 mm |

## 2 Specifications · physical

| FEATURE | SPECIFICATION |
|---|---|
| Substrate | Aluminum / LDPE / aluminum composite, 6 mm |
| Weight | 6.6 kg/m² (300 × 200: ≈ 0.4 kg · 800 × 600: ≈ 3.2 kg) |
| Pattern | Direct UV print, matte, black on white; optical density ≥ 1.6 |
| Flatness | ≤ 0.5 mm per meter, board supported on three points |
| Mounting | 4 × M4 threaded inserts on the rear (300 × 200 and larger); ¼″ fixture CT-ACC-M-FIX14 clamps any size |

## 3 Specifications · environment

| FEATURE | SPECIFICATION |
|---|---|
| Operating temperature | −50 to +80 °C |
| Thermal expansion | ≈ 24 µm/(m·°C) |
| Storage | −20 to +60 °C, dry |
| Outdoor lifetime | up to 3 years (UV-stable print) |
| Cleaning | water or ≤ 70 % isopropyl alcohol, soft cloth |

## 4 Deliverables

| INCLUDED | | SERVICE | |
|---|---|---|---|
| Measurement | Every board is measured before it ships (dimensional, flatness, visual) | Lead time | Made to order · ships in 2–3 weeks |
| Files | `.opencv.json`, `.kalibr.yaml`, `.ros2.yaml`, `.dxf` | Options | Logo print (-L) · hard case (CT-ACC-M-CSE400) · custom NRE |
| Documents | this datasheet, CT-UM-GEN user manual | | |

## 5 Compliance & customs

| TOPIC | STATEMENT |
|---|---|
| RoHS | RoHS-compliant materials. |
| HS code | `9031 80` Not a toy. not for food contact. |

## 6 Revision history

| REV | DATE | CHANGE |
|---|---|---|
| A | `2026-10` | First issue, planning values |

## 7 Disclaimer

Values are nominal unless stated as measured.